INCH-POUND

MIL-DTL-27189E
19 February 2008
Supersedes
MIL-I-27189D
15 October 1993

DETAIL SPECIFICATION

INDICATOR, PRESSURE, OIL

This document is inactive for new design after 24 March 1999.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers hermetically sealed, remote indicating, voltage ratio type integrally lighted, oil pressure indicators.

1.2 Classification. Indicators shall be of the following types, as specified (see 6.2):

MS24576-1	Red lighting
MS24576-2	White lighting
MS24576-3	White lighting with multilayer coated antireflection cover glass

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL STANDARDS

FED-STD-595	Color Used in Government Procurement (Color Chips 37038 and 37875).

Comments, suggestions, or questions on this document should be addressed to: Oklahoma City Air Logistics Center/ENEP, 3001 Staff Drive, Suite 2AG-68A, Tinker AFB, OK 73145-3036 or emailed to ocalc.enrs.sma@tinker.af.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST database at http://assist.daps.dla.mil

AMSC N/A FSC 6620

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DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-C-675	Coating of Glass Optical Elements (Anti-reflection)
MS24576	Indicator, Pressure, Oil
MIL-PRF-25435	Transmitter, Pressure, Aircraft, General Specification For
MIL-I-25437	Indicators, Aircraft, Pressure, Voltage Ratio Type, General Specification For
MIL-L-25467	Lighting, Integral, Red, Aircraft Instrument, General Specification For
MIL-L-27160	Lighting, Instrument, Integral, White, General Specification For
MS33639	Cases, Instruments, Clamp-mounted, Aircraft
MIL-PRF-38230	Transmitter, Pressure, Oil,

DEPARTMENT OF DEFENSE HANDBOOKS

MIL-HDBK-781	Reliability Test Methods, Plans and Environments for Engineering, Development Qualification, and Production

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or http://assist.daps.dla.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. The indicators furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified manufacturers list before contract award (see 4.3 and 6.3).

3.2 General specification. The requirements of MIL-I-25437 apply as requirements of this specification with the exceptions and additions specified herein. When the two specifications conflict, this specification shall govern.

3.3 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.4 Item requirements. The individual item requirements shall be as specified herein and in accordance with the applicable specification. In the event of any conflict between the requirements of this specification and the specification sheet, the later shall govern.

3.5 Design, construction, and physical dimensions.

3.5.1 Indicator. The indicator shall be designed for use with a TRU-66/A transmitter conforming to MIL-PRF-38230.

3.5.2 Case. The case dimensions shall conform to MS33639 the 2-inch size. The length of the case shall not exceed 2.75 inches.

3.5.2.1 Cover Glass.

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3.5.2.1.1 Perfect Reflectance (MS24576-3). All reflecting glass surfaces of the MS24576-3 indicator shall be provided with a reflection reducing coating which meets the requirements of MIL-C-675 with the exception of the materials use, in addition to withstanding the environmental conditions specified herein. The following reflectance tolerances shall apply:

Angle of Incidence	Wavelength Millimicrons	Percent Reflectance
0°	450-675	0.6 absolute
0°	425-700	0.5 average
30°	400-625	1.0 absolute
30°	425-700	0.5 average

3.5.3 Reliability. The indicator shall have a minimum acceptable mean-time-between-failures (MTBF) of 2,000 hours based upon a discrimination ratio of 1.5 and confidence factor of 90 percent.

3.5.3.1 Longevity. The indicator shall have an operating life span of 3,000 hours.

3.6 Performance.

3.6.1 Tolerances. The indicator shall operate within the tolerances specified inTables I and II.

TABLE I. Temperature Scale Error.

Test Pressure, PSI	Room Temperature, PSI	Scale Error High and Low Temp Exposure High and Low Temperature, PSI
0	1.5	3.00
10	1.5	3.00
20	1.5	3.00
30	1.5	3.00
40	1.5	3.00
50	1.5	3.00
60	1.5	3.00
70	1.5	3.00
80	1.5	3.00
90	1.5	3.00
100	1.5	3.00

TABLE II. Position, Friction, and Vibration Error.

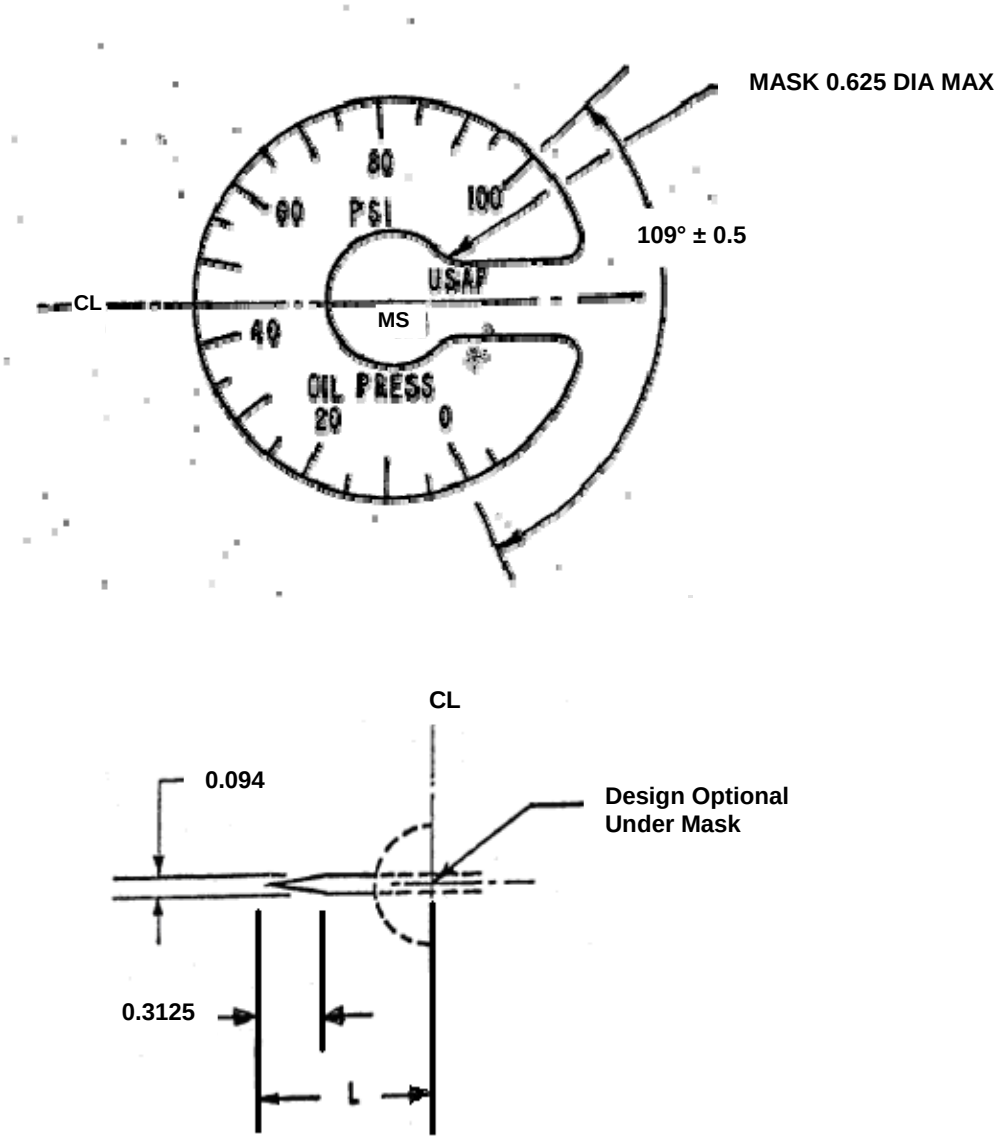
Test	Tolerance +PSI
Position error at 50 psi	1.5
Friction	
Room temperature	1.5
High temperature	2.0
Low temperature	2.25
Vibration	
Pointer oscillation	2.0
Pointer variation	1.5

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3.7 Lighting. The indicators shall be integrally lighted and shall meet all requirements specified herein. The MS24576-1 indicator shall be lighted in accordance with MIL-L-25467 and the MS24576-2 and MS24576-3 indicators shall be lighted in accordance with MIL-L-27160.

3.8 Pointer. The pointer shall be in accordance with figure 1 except that the length shall be such that the tip will extend into the scale a distance equal to approximately one-third the length of the shortest graduation. The pointer shall be suitable for integral lighting.

3.8.1 Pointer finish. The lighted portion of the pointer shall be finished in accordance with FED-STD-595, color No. 37875.



Dimensions In Inches Unless Otherwise Specified, Tolerances: Decimal 0.016

FIGURE 1. Oil Pressure Indicator

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3.9 Dial. The dial shall conform to figure 1 and shall be of material suitable for integral lighting. A minimum diameter of 1.75 inches shall be maintained across the outside of the graduations. The graduation for 45 psi shall be located on the horizontal centerline at the left side of the dial.

3.9.1 Markings.

3.9.1.1 White. The following markings shall be finished in durable white in accordance with FED-STD-595, color No. 37875. The dimensions shall be as specified.

	Height or Length Inch +0.016	Width of Line or Graduation Inch +0.005
Numerals 0, 20, 40, 60, 80, and 100	0.157	0.025
10-pound graduations	0.188	0.031
5-pound graduations	0.094	0.020
Lettering OIL PRESS	0.125	0.025
Lettering PSI	0.125	0.025

3.9.1.2 Black. The markings USAF and MS24576 (*) (applicable dash No.) shall be permanently and legibly marked on the mask in letters 0.062 inch in height. Unless otherwise specified, these, all other markings, and backgrounds shall be finished in durable dull black in accordance with FED-STD-595, color No. 37038.

3.10 Electrical connector. The indicator shall be provided with an electrical connector, Deutsch Company part No. DM5600-7P, or equal, to mate with Deutsch Company part No. DM9700-7S, or equal. The connector shall be furnished with a cap or cover and gasket to prevent penetration of water vapor.

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3.11 Electrical wiring. The internal and external wiring shall conform to Figure 2. The internal wiring shall be color coded and shall be connected to the receptacle pins as shown.

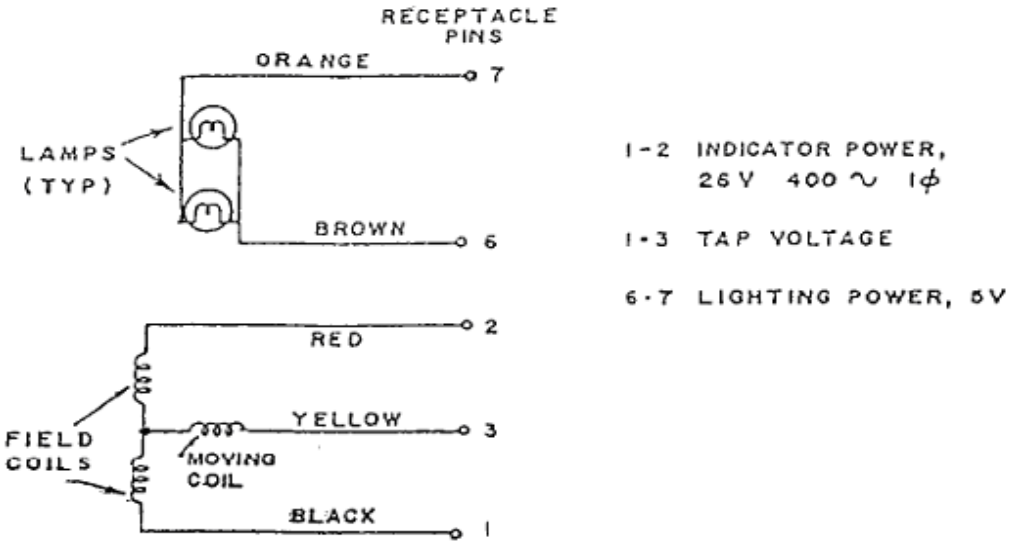


FIGURE 2. Wiring Diagram.

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3.12 Electrical characteristics. The current drawn by the indicator shall not exceed 100 ma with pins 3, 6, and 7 open circuited. The power consumption of the indicator, exclusive of lamps, shall not exceed 1 w.

3.1 Weight. The weight of the indicator shall not exceed 0.50 pound.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specific herein are classified as follow:

a. Qualification inspection (see 4.3)
b. Quality conformance inspection (see MIL-I-25437 section 4.4)

4.1.1 General specification. The inspection and testing of indicators shall be in accordance with MIL-I-25437 and as specified herein. The indicators shall operate within the tolerances specified in 3.6.1.

4.2 Qualification inspection.

4.2.1 Test samples. Ten indicators are required for reliability testing. Three indicators are required for all other tests. All indicators shall be representative of the production configuration.

4.2.2 Qualification tests. The Qualification tests shall consist of all the tests specified herein and in MIL-I-25437.

4.3 Test methods.

4.3.1 Lighting. The MS24576-1 indicator shall be subjected to lighting tests in accordance with MIL-L-25467. MS24576-2 and MS24576-3 indicators shall be subjected to lighting tests in accordance with MIL-L-27160.

4.3.2 Reliability.

4.3.2.1 Demonstration. Unless otherwise specified, ten indicators shall be subjected to the reliability test specified in 4.3.2.2 to demonstrate a minimum required MTBF of 2,000 hours with a confidence level of 90% and a discrimination ratio of 1.5. See MIL-HDBK-781 for guidance. The total test time, number of failures permitted, and accept/continue/reject criteria shall be in accordance with figure 3.

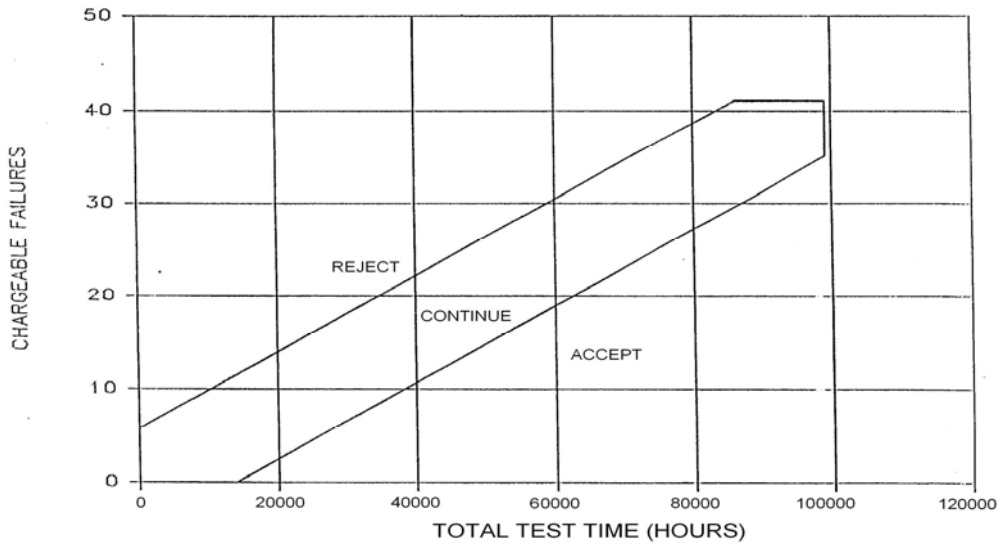


FIGURE 3. Reliability Accept/Reject

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4.3.2.2 Test. The test equipment and fixtures shall include provisions for vibration and temperature cycling at the periodic intervals specified herein. During this test, simulated pressure transmitter signals shall be applied to drive the indicator reading from 0 to 85 psi and back at a rate of 2 to 3 cycles per minute. The test shall begin when rated power is applied to the indicator. The test shall include at least 480 temperature cycles of operation. During each cycle, power shall be turned off and shall remain off for a period of 10 minutes. For at least 10 minutes out of every hour of ON time, the indicator shall be vibrated in accordance with MIL-STD-810 Method 514 using the spectra shown in figure 4. The temperature cycles shall be conducted in the following sequence: 4 hours at -20°C, 14 minutes for temperature change to +50° C at a minimum rate of 5° C per minute; 4 hours at +50° C, 14 minutes for temperature change to -20° C. In case of malfunction, the malfunction shall be recorded and the test interrupted for repairs. Upon completion of each 250 hours of cycling, the indicator shall be checked for scale error. The scale error readings shall not exceed 150 percent of the room temperature tolerances in table I. No recalibration shall be performed on the units during this test. The test may be interrupted for weekend shutdown.

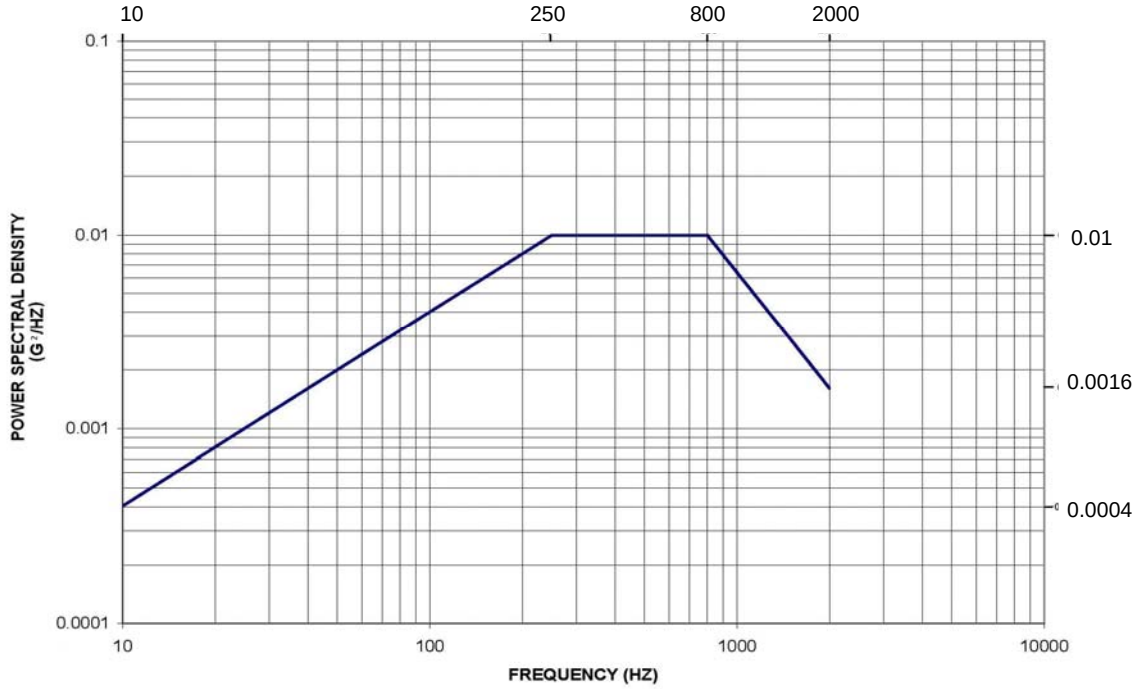


FIGURE 4. Random Vibration Spectra

4.3.2.3 Longevity. At the completion of the reliability test, three of the indicators shall continue to be tested for a total of 3000 hours each. The test time accumulated during the reliability test shall be applied to this test. The accept-reject criteria are not applicable.

4.3.3 Compatibility. One indicator shall be submitted to the preparing activity for a compatibility test. The indicator shall be connected to a variety of transmitters which meet the requirements of MIL-PRF-38230. The indicator and transmitter shall be connected per figure 2. The scale error test of MIL-PRF-25435 shall be performed at room temperature. The tolerance at each test point shall be the sum of the scale error tolerances of MIL-PRF-38230 and table I of this document at room temperature.

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PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order. When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The indicators covered by this specification are intended for use in conjunction with TRU-66/A pressure transmitter in indicating remotely the pressure in oil systems of aircraft engines.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification
b. Type of indicator required (see 1.2)
c. Packaging requirements (see 5.1)

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL-27189 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Oklahoma City Air Logistics Center/ENEP, 3001 Staff Drive, Suite 2AG-68A, Tinker AFB, OK 73145-3036 or emailed to ocalc.enrs.sma@tinker.af.mil

6.4 Subject term (key word) listing:

Lighted
MIL-PRF-38230
Variable reluctance
Voltage ratio
0-100 PSI

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodian:
Air Force - 71

Preparing activity:
Air Force - 71

Review activities
Air Force - 99
DLA - GS

(Project 6620-2007-005)

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